Variable Selection for Wasserstein Robust Hypothesis Testing

1 Problem formulation

Throughout this paper, we consider the variable selection problem for robust hypothesis testing. Let $[D] = \{1, \ldots, D\}$ and d be a positive integer denoting the sparsity budget. Given a detector $f : \mathbb{R}^D \to \mathbb{R}$, we define the worst-case risk as

$$R(f) = \sup_{\mathbb{P}_0 \in \mathcal{P}_0, \mathbb{P}_1 \in \mathcal{P}_1} \left\{ \mathbb{E}_{\mathbb{P}_0} \ell(-f(X)) + \mathbb{E}_{\mathbb{P}_1} \ell(f(X)) \right\}, \tag{1}$$

where $\mathcal{P}_k, k = 0, 1$ denote the Wasserstein distance-based ambiguity sets, and $\ell(\cdot)$ is the generating function that is nonnegative-valued, non-decreasing, convex, and satisfies $\ell(0) = 1, \lim_{t \to -\infty} \ell(t) = 0$. For $k = 0, 1$, assume we have n_k available observations in the group k, which are denoted as $\{x_i^{(k)}\}_{i=1}^{n_k}$. We denote the corresponding empirical distribution as $\widehat{\mathbb{P}}_k = \frac{1}{n_k} \sum_{i=1}^{n_k} \delta_{x_i^{(k)}}$, and construct the Wasserstein ambiguity set as

$$\mathcal{P}_k = \left\{ \mathbb{P}_k : \mathcal{W}_1(\widehat{\mathbb{P}}_k, \mathbb{P}_k) \le \rho_k \right\}.$$

Here, $\mathcal{W}_1(\cdot, \cdot)$ refers to the 1-Wasserstein distance equipped with the ℓ_1-norm.

For $S \subseteq [D]$, define the collection of detectors whose input only depends on coordinates belonging to S:

$$\mathcal{F}_S = \left\{ f : \mathbb{R}^D \to \mathbb{R}, f(x) = g(x[S]) \text{ for some measurable function } g : \mathbb{R}^{|S|} \to \mathbb{R} \right\}.$$

Note that for a given vector $x \in \mathbb{R}^D$, we use $x[i]$ to denote its i-th coordinate, and $x[S]$ to denote the concatenation of its coordinates whose indices belong to S. Our proposed variable selection problem can be formulated as the following combinatorial optimization problem

$$\min_{S \subseteq [D], |S| \le d} \min_{f \in \mathcal{F}_S} R(f). \tag{2}$$

2 Convex formulation

For $S \subseteq [D]$, denote $V(S) = \min_{f \in \mathcal{F}_S} R(f)$ and define

$$\mathcal{X} = \{x_i^{(0)} : i \in [n_0]\} \cup \{x_j^{(1)} : j \in [n_1]\}.$$

Assume $n_0, n_1 \ge 1$, $\rho_0, \rho_1 > 0$, and that ℓ is finite on $\mathbb{R}$. Each target $y \in \mathcal{X}$ receives a detector score g_y, and $g = (g_y)_{y \in \mathcal{X}}$. For masses $a, b \ge 0$, let $\psi(a, b) = \inf_{t \in \mathbb{R}} \{a\ell(-t) + b\ell(t)\}$ be the smallest loss at a point that carries mass a of class 0 and b of class 1; it is the auxiliary function of [2] in homogeneous form. The next lemma is [2, Proposition 1 and Theorem 3] applied to the projected observations: projection contracts the ℓ_1 transport cost, and every projected transport lifts at the same cost by keeping each source's unselected coordinates.

Lemma 2.1 (Least favorable distributions)**.**

$$\begin{aligned} V(S) = \max_{p, \gamma \ge 0} \quad & \sum_{y \in \mathcal{X}} \psi\big(p_0(y), p_1(y)\big) \\ \textit{subject to} \quad & \sum_{y \in \mathcal{X}} \gamma_{i,y}^{(k)} = \frac{1}{n_k}, && i \in [n_k],\ k = 0, 1, \\ & p_k(y) = \sum_{i=1}^{n_k} \gamma_{i,y}^{(k)}, && y \in \mathcal{X},\ k = 0, 1, \\ & \sum_{i=1}^{n_k} \sum_{y \in \mathcal{X}} \gamma_{i,y}^{(k)} \|x_i^{(k)}[S] - y[S]\|_1 \le \rho_k, && k = 0, 1. \end{aligned} \tag{3}$$

Here $\gamma_{i,y}^{(k)}$ is the mass transported from $x_i^{(k)}[S]$ to $y[S]$, and $p_k(y)$ is the mass of class k at that target. Let Γ_S denote the feasible transports in Eq. (3) after eliminating p. Since $\psi(a,b)$ is an infimum over scores, the objective equals $\inf_g \sum_k \sum_{i,y} \gamma_{i,y}^{(k)} \ell((-1)^{k+1} g_y)$. This sum is convex in g and linear in γ, and Γ_S is compact and convex, so Sion's minimax theorem [3] lets us minimize over g outside:

$$V(S) = \inf_g \max_{\gamma \in \Gamma_S} \sum_{k=0}^{1} \sum_{i,y} \gamma_{i,y}^{(k)} \ell((-1)^{k+1} g_y).$$

For fixed g, the inner maximum splits into one linear program per class: the worst expectation of $y \mapsto \ell((-1)^{k+1} g_y)$ over transports of $\widehat{\mathbb{P}}_k$ with budget ρ_k. Leaving every observation in place is feasible, so linear programming duality gives the familiar Wasserstein dual

$$\min_{\lambda_k \geq 0} \ \rho_k \lambda_k + \frac{1}{n_k} \sum_{i=1}^{n_k} \max_{y \in \mathcal{X}} \left\{ \ell((-1)^{k+1} g_y) - \lambda_k \| x_i^{(k)}[S] - y[S] \|_1 \right\}.$$

Naming the inner maximum $a_i^{(k)}$ and minimizing over g gives

$$\begin{aligned} V(S) = \min_{g,a,\lambda} \quad & \frac{1}{n_0} \sum_i a_i^{(0)} + \frac{1}{n_1} \sum_j a_j^{(1)} + \rho_0 \lambda_0 + \rho_1 \lambda_1 \\ \text{subject to} \quad & \ell(-g_y) \leq a_i^{(0)} + \lambda_0 \| x_i^{(0)}[S] - y[S] \|_1, \qquad i \in [n_0],\ y \in \mathcal{X}, \\ & \ell(g_y) \leq a_j^{(1)} + \lambda_1 \| x_j^{(1)}[S] - y[S] \|_1, \qquad j \in [n_1],\ y \in \mathcal{X}, \\ & g \in \mathbb{R}^{\mathcal{X}}, \quad a^{(k)} \in \mathbb{R}^{n_k}, \quad \lambda_k \geq 0, \qquad k = 0, 1. \end{aligned} \tag{4}$$

Eq. (4) is a finite convex program, and it attains its minimum: the constraints with $y = x_i^{(k)}$ give $a_i^{(k)} \geq 0$, so an objective value at most B forces $a_i^{(k)} \leq n_k B$ and $\lambda_k \leq B/\rho_k$; the constraints then bound $\ell(\pm g_y)$, hence g, since $\ell(t) \to \infty$ as $t \to \infty$ (ℓ is convex with $\ell(0) = 1 > \lim_{t \to -\infty} \ell(t)$).

A detector is recovered from an optimal (g, a, λ) by extending the scores off the samples. For $\xi \in \mathbb{R}^{|S|}$, let $E_k(\xi) = \min_{i \in [n_k]} \{ a_i^{(k)} + \lambda_k \| x_i^{(k)}[S] - \xi \|_1 \}$. The constraints of Eq. (4) say exactly that $\ell(-g_y) \leq E_0(y[S])$ and $\ell(g_y) \leq E_1(y[S])$, so we require the extension to satisfy

$$\ell(-g(\xi)) \leq E_0(\xi), \qquad \ell(g(\xi)) \leq E_1(\xi), \qquad \xi \in \mathbb{R}^{|S|}. \tag{5}$$

This is the analogue, for a general ℓ, of the interval that [4, Theorem 1] gives for randomized tests. Coincident projected targets have identical constraints, so $g(y[S]) = g_y$ is well defined and satisfies Eq. (5). For any other ξ, let i and j be the smallest indices attaining $E_0(\xi)$ and $E_1(\xi)$, let $r_0 = \| x_i^{(0)}[S] - \xi \|_1$, $r_1 = \| x_j^{(1)}[S] - \xi \|_1$ and $b = \| x_i^{(0)}[S] - x_j^{(1)}[S] \|_1$, and interpolate between the two anchors: $g(\xi) = (1 - \tau) g_{x_i^{(0)}} + \tau g_{x_j^{(1)}}$ with $\tau = \min\{1, r_0/b\}$ if $b > 0$ and $\tau = 0$ otherwise. The constraints of Eq. (4) at source i give $\ell(-g_{x_i^{(0)}}) \leq a_i^{(0)}$ and $\ell(-g_{x_j^{(1)}}) \leq a_i^{(0)} + \lambda_0 b$; by convexity, $\ell(-g(\xi)) \leq a_i^{(0)} + \lambda_0 \tau b \leq E_0(\xi)$, since $\tau b \leq r_0$. Likewise $\ell(g(\xi)) \leq a_j^{(1)} + \lambda_1 (1 - \tau) b \leq E_1(\xi)$, since $(1 - \tau) b \leq r_1$ by the triangle inequality.

The detector $f(x) = g(x[S])$ is Borel and attains $R(f) = V(S)$. Indeed, E_k is λ_k-Lipschitz for $\| \cdot \|_1$ and $E_k(x_i^{(k)}[S]) \leq a_i^{(k)}$. Projection does not increase $\mathcal{W}_1$, so by Kantorovich–Rubinstein duality every $\mathbb{P}_k \in \mathcal{P}_k$ satisfies $\mathbb{E}_{\mathbb{P}_k} E_k(X[S]) \leq \frac{1}{n_k} \sum_i a_i^{(k)} + \lambda_k \rho_k$. With Eq. (5), summing over k bounds $R(f)$ by the value of Eq. (4), which is $V(S)$; the reverse inequality holds because $f \in \mathcal{F}_S$.

To optimize the coordinate selection, define $z = \mathbf{1}_S$ and $\mathcal{Z}_d = \{ z \in \{0,1\}^D : \mathbf{1}^\top z \leq d \}$. For $\gamma = (\gamma^{(0)}, \gamma^{(1)})$ with $\gamma^{(k)} \in \mathbb{R}_+^{n_k \times |\mathcal{X}|}$, let $p_k(y) = \sum_i \gamma_{i,y}^{(k)}$ as in Eq. (3) and define the objective of Eq. (3)

$$H(\gamma) = \sum_{y \in \mathcal{X}} \psi\big(p_0(y), p_1(y)\big).$$

Since ψ is an infimum of linear functions with nonnegative coefficients, H is concave, nondecreasing, and homogeneous: $H(s\gamma) = sH(\gamma)$ for $s \geq 0$. Define the average of the two class masses and the

transport cost in coordinate m, respectively, by

$$M(\gamma) = \frac{1}{2}\sum_{k=0}^{1}\sum_{i=1}^{n_k}\sum_{y\in\mathcal{X}}\gamma_{i,y}^{(k)},$$
$$c_k(\gamma)[m] = \sum_{i=1}^{n_k}\sum_{y\in\mathcal{X}}\gamma_{i,y}^{(k)}|x_i^{(k)}[m] - y[m]|, \qquad k = 0,1,\ m \in [D].$$

For a scalar $t \in \mathbb{R}$, write $t_+ = \max\{t, 0\}$. The next result writes the reciprocal of the selection objective as a minimum of functions affine in z.

Proposition 2.2 (Convex selection function)**.** *Let $\mathcal{D}$ be the set of pairs (γ, μ) with $\gamma \geq 0$ and $\mu \in \mathbb{R}_+^D$ such that*

$$\sum_{y\in\mathcal{X}}\gamma_{i,y}^{(k)} = \frac{M(\gamma)}{n_k}, \quad i \in [n_k], \qquad \sum_{m=1}^{D}\big(c_k(\gamma)[m] - \rho_k\mu_m\big)_+ \leq \rho_k M(\gamma), \qquad k = 0,1.$$

For $z \in [0,1]^D$, define $F(z) > 0$ by

$$\frac{1}{F(z)} = \min\big\{M(\gamma) + \mu^\top z :\ (\gamma,\mu) \in \mathcal{D},\ H(\gamma) \geq 1\big\}. \tag{6}$$

Then the minimum is attained, F is convex and nonincreasing with $F \leq 2$, and

$$F(\mathbf{1}_S) = V(S), \qquad \min_{|S|\leq d} V(S) = \min_{z\in\mathcal{Z}_d} F(z). \tag{7}$$

The row constraints make each class's total mass equal to $M(\gamma)$; this common mass may vary, whereas each class has unit mass in Eq. (3). Because H, M and $\mathcal{D}$ scale together, $F(z)$ is the largest ratio $H(\gamma)/(M(\gamma) + \mu^\top z)$ over $\mathcal{D}$. The variable μ_m is a price: paying $\mu_m z_m$ relieves each class of $\rho_k\mu_m$ of transport cost in coordinate m. At $z = \mathbf{1}_S$ the unselected coordinates are relieved for free, which leaves the budget of Eq. (3) on S alone. Appendix A proves the proposition directly from Lemma 2.1. Minimizing F over $\mathcal{Z}_d$ therefore solves the original problem; replacing $\mathcal{Z}_d$ by its continuous relaxation gives a lower bound.

3 An outer-approximation approach

We use outer approximation [1] on the concave function $1/F$. For fixed z, Eq. (6) minimizes a linear function over a convex set, since $\mathcal{D}$ is convex and H is concave. For an optimizer (γ^t, μ^t) at z^t, define

$$c_t(z) = M(\gamma^t) + (\mu^t)^\top z. \tag{8}$$

The feasible set in Eq. (6) does not depend on z, so $1/F(z) \leq c_t(z)$ for every $z \in [0,1]^D$, with equality at z^t. Equivalently, $F(z) \geq 1/c_t(z)$; as $t \mapsto 1/t$ is convex, this bound lies above the tangent of F at z^t. At a binary $z^t = \mathbf{1}_{S^t}$ no extra solve is needed: if $\gamma^\star$ is a least favorable transport of Eq. (3) at S^t, Appendix A shows that $\gamma^t = \gamma^\star/V(S^t)$ with $\mu_m^t = \max_k c_k(\gamma^\star)[m]/(\rho_k V(S^t))$ for $m \notin S^t$ and $\mu_m^t = 0$ for $m \in S^t$ is optimal. After t evaluations, we maximize the accumulated upper bounds on $1/F$ to obtain the next selection:

$$\begin{aligned}(z^{t+1}, \theta_t) \in \operatorname*{arg\,max}_{z\in\mathcal{Z}_d,\,\theta} \quad & \theta \\ \text{subject to} \quad & \theta \leq c_s(z), \qquad s = 1,\dots,t.\end{aligned} \tag{9}$$

The value $L_t = 1/\theta_t$ is a lower bound on OPT $= \min_{z\in\mathcal{Z}_d} F(z)$, and the best evaluated value $U_t = \min_{s\leq t} F(z^s)$ is an upper bound.

Algorithm 1 alternates between evaluating a selection and solving Eq. (9). It retains the best evaluated detector and stops when $U_t - L_t \leq \varepsilon$. With exact solves, if $z^{t+1} = z^s$ for some $s \leq t$, then

$$\theta_t \leq c_s(z^{t+1}) = 1/F(z^s) \leq 1/U_t,$$

so $L_t \geq U_t$ and the stopping condition holds. Each nonterminal iteration therefore produces a new selection, and at most $|\mathcal{Z}_d|$ selections are evaluated. Since F is nonincreasing, the budget can also be imposed as $\mathbf{1}^\top z = d$ when $d \leq D$.

Algorithm 1 Outer approximation for variable selection

Require: Samples $x_i^{(k)}$, radii $\rho_k > 0$, budget d, tolerance $\varepsilon > 0$, initial $z^1 \in \mathcal{Z}_d$

$U \leftarrow +\infty$, $t \leftarrow 1$
loop
  $S^t \leftarrow \{m : z_m^t = 1\}$
  Solve Eq. (4) at S^t for (g^t, a^t, λ^t), value $v_t = V(S^t)$, and a least favorable transport of Eq. (3)
  Form (γ^t, μ^t) from it and c_t by Eq. (8)
  if $v_t < U$ **then**
    $U \leftarrow v_t$; store $S^\star \leftarrow S^t$ and $(g^\star, a^\star, \lambda^\star) \leftarrow (g^t, a^t, \lambda^t)$
  end if
  Solve Eq. (9) with $c_1, \ldots, c_t$ for (z^{t+1}, θ_t); $L_t \leftarrow 1/\theta_t$
  if $U - L_t \leq \varepsilon$ **then**
    Extend $g^\star$ to $\mathbb{R}^{|S^\star|}$ by Eq. (5)
    return $S^\star$ and $f^\star(x) = g^\star(x[S^\star])$
  end if
  $t \leftarrow t + 1$
end loop

When a solver stops early, feasible solutions of Eq. (4) give upper bounds, and every $(\gamma, \mu) \in \mathcal{D}$ with $H(\gamma) > 0$ gives the valid cut $(M(\gamma) + \mu^\top z)/H(\gamma) \geq 1/F(z)$. Take θ_t as the solver's global upper bound for Eq. (9).

A Proof of Proposition 2.2

A.1 Equality at binary selections

Fix S. We first note $V(S) > 0$. Convexity of ℓ gives $\ell(t) + \ell(-t) \geq 2\ell(0) = 2$, hence $\psi(a, b) \geq 2\min\{a, b\}$. Moving a small mass $\varepsilon > 0$ of $x_1^{(0)}$ onto the target $x_1^{(1)}$ is feasible in Eq. (3) and puts masses at least ε and $1/n_1$ on that target, so $V(S) \geq 2\min\{\varepsilon, 1/n_1\} > 0$.

Upper bound on $1/F(\mathbf{1}_S)$. The maximum in Eq. (3) is attained, because Γ_S is compact and H, an infimum of linear functions, is upper semicontinuous. Let $\gamma^\star$ attain it, so $M(\gamma^\star) = 1$, $H(\gamma^\star) = V(S)$ and $\sum_{m \in S} c_k(\gamma^\star)[m] \leq \rho_k$. Set $\mu_m = 0$ for $m \in S$ and $\mu_m = \max_k c_k(\gamma^\star)[m]/\rho_k$ for $m \notin S$. The terms with $m \notin S$ then vanish, and

$$\sum_m \big(c_k(\gamma^\star)[m] - \rho_k \mu_m\big)_+ = \sum_{m \in S} c_k(\gamma^\star)[m] \leq \rho_k M(\gamma^\star).$$

Thus $(\gamma^\star, \mu) \in \mathcal{D}$. The constraints of $\mathcal{D}$ are invariant under scaling $(\gamma, \mu) \mapsto (s\gamma, s\mu)$ with $s > 0$, and H and M scale by s. The pair $(\gamma^\star, \mu)/V(S)$ is therefore feasible in Eq. (6) with $H = 1$, and its objective is $(M(\gamma^\star) + \mu^\top \mathbf{1}_S)/V(S) = 1/V(S)$. Hence $1/F(\mathbf{1}_S) \leq 1/V(S)$.

Lower bound on $1/F(\mathbf{1}_S)$. Let $(\gamma, \mu) \in \mathcal{D}$ with $H(\gamma) \geq 1$, and let $s = M(\gamma) + \mu^\top \mathbf{1}_S$. Taking $t = 0$ in the definition of ψ gives $\psi(a, b) \leq a + b$, hence $H(\gamma) \leq 2M(\gamma)$, so $M(\gamma) \geq 1/2$ and $s > 0$. Let $\iota_{i,y}^{(k)} = n_k^{-1} \mathbf{1}_{\{y = x_i^{(k)}\}}$ be the transport that leaves each observation in place; then $M(\iota) = 1$ and $c_k(\iota) = 0$. Rescale γ and complete its mass with ι:

$$\gamma' = \frac{\gamma}{s} + \Big(1 - \frac{M(\gamma)}{s}\Big)\iota.$$

Since $M(\gamma) \le s$, $\gamma' \ge 0$, and its rows sum to $1/n_k$. For each $m \in S$, $c_k(\gamma)[m] \le (c_k(\gamma)[m] - \rho_k\mu_m)_+ + \rho_k\mu_m$. Summing over $m \in S$, bounding the positive parts by their sum over all m, and using the cost constraint of $\mathcal{D}$,

$$\sum_{m\in S} c_k(\gamma')[m] = \frac{1}{s}\sum_{m\in S} c_k(\gamma)[m] \le \frac{\rho_k M(\gamma) + \rho_k \mu^\top \mathbf{1}_S}{s} = \rho_k.$$

Hence $\gamma' \in \Gamma_S$. As H is nondecreasing and homogeneous, $V(S) \ge H(\gamma') \ge H(\gamma/s) = H(\gamma)/s \ge 1/s$. Every feasible point of Eq. (6) at $\mathbf{1}_S$ therefore has objective $s \ge 1/V(S)$, so $1/F(\mathbf{1}_S) \ge 1/V(S)$. Together, $F(\mathbf{1}_S) = V(S)$, and $(\gamma^\star, \mu)/V(S)$ attains the minimum; this is the optimizer used in Section 3.

A.2 Attainment and convexity

Fix $z \in [0,1]^D$. The feasible set of Eq. (6) is nonempty by Appendix A.1, and every feasible point has $M(\gamma) + \mu^\top z \ge M(\gamma) \ge 1/2$, so $1/F(z) \ge 1/2$ and $F(z) \le 2$. Take a minimizing sequence (γ^j, μ^j). Its objective values are bounded, so $M(\gamma^j)$ and hence γ^j are bounded. If μ^j_m exceeds $\max_k c_k(\gamma^j)[m]/\rho_k$, lowering it to this value keeps the terms $(c_k(\gamma^j)[m] - \rho_k\mu^j_m)_+$ at zero and does not increase the objective, so we may take μ^j bounded as well. A subsequence converges; its limit is feasible, because $\mathcal{D}$ is closed and H is upper semicontinuous, and it attains the minimum.

The set $\mathcal{D} \cap \{H \ge 1\}$ is convex, because the constraints of $\mathcal{D}$ are linear equations and a sum of positive parts of affine functions bounded by a linear function, and H is concave. This set does not depend on z, and the objective is affine in z with slope $\mu \ge 0$. Hence $1/F$, a pointwise minimum of such functions, is concave, nondecreasing and positive. Since $t \mapsto 1/t$ is convex and decreasing on $(0,\infty)$, F is convex and nonincreasing. With Appendix A.1, the correspondence $S \leftrightarrow \mathbf{1}_S$ gives Eq. (7). □

References

[1] Dimitris Bertsimas, Ryan Cory-Wright, and Jean Pauphilet. A unified approach to mixed-integer optimization problems with logical constraints. *SIAM Journal on Optimization*, 31(3):2340–2367, 2021.

[2] Rui Gao, Liyan Xie, Yao Xie, and Huan Xu. Robust hypothesis testing using Wasserstein uncertainty sets. In *Advances in Neural Information Processing Systems*, volume 31. Curran Associates, Inc., 2018.

[3] Maurice Sion. On general minimax theorems. *Pacific Journal of Mathematics*, 8(1):171–176, March 1958.

[4] Liyan Xie, Rui Gao, and Yao Xie. Robust hypothesis testing with Wasserstein uncertainty sets, 2021. arXiv:2105.14348.